

<page_start no = 1>

<sol_start id=1>

1. The light beam will be real and inverted.

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<sol_start id=2>

2. $f = \frac{R}{2}$ [Radius of curvature = 50]

$$f = \frac{50}{2}$$

$$f = 25$$

focal length = 25

It will be a convex mirror

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4. $h_2 \Rightarrow 2 \text{ cm}$

$f \Rightarrow 10 \text{ cm}$

$U \Rightarrow 15 \text{ cm}$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{U}$$

$$\frac{1}{10} = \frac{1}{v} + \frac{1}{15}$$

$$\frac{1}{10} - \frac{1}{15} = \frac{1}{v}$$

$$\frac{3}{30} - \frac{2}{30} = \frac{1}{v}$$

$$\frac{1}{30} = \frac{1}{v}$$

$v = +30 \rightarrow$ (30 cm from the mirror is the position of image)

$$m = \frac{+v}{U}$$

$$m = \frac{\cancel{30}}{\cancel{15}}$$

Magnification $\Rightarrow 2$

Size is magnified because $m = 2$ $m > 1$

Nature $\Rightarrow$ virtual and erect.

<sol_end>

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